

MONTERREY INTL, MexicoWMO: **763943**Lat: **25.778N**Lon: **100.107W**Elev: **1278**StdP: **14.03**Time Zone: **-6.00 (MEX)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.3	40.8	16.0	12.9	59.2	22.6	17.8	59.9	25.5	63.1	22.9	61.3	7.1	330	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	101.9	73.0	100.0	72.8	98.3	72.8	79.4	93.5	78.4	91.8	77.6	90.5	12.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	140.5	84.6	75.2	138.5	84.2	73.8	131.9	82.6	44.0	93.1	43.0	93.7	42.0	89.3	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.9	17.4	32.3	106.4	3.8	3.3	29.6	108.7	27.3	110.6	25.2	112.4	22.4	114.8		
(5)			29.0	81.9	3.9	3.5	26.2	84.4	23.9	86.4	21.7	88.4	18.9	90.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.7	59.3	64.0	69.6	75.8	80.8	84.3	84.7	84.9	80.4	74.0	66.1	59.8		
	DBStd	11.31	8.65	8.88	7.88	6.78	5.54	3.43	3.43	2.88	4.28	6.39	7.48	8.21		
	HDD50	50	24	8	1	0	0	0	0	0	0	0	3	14		
	HDD65	679	215	117	48	8	0	0	0	0	0	12	76	203		
	CDD50	8689	312	399	608	774	954	1028	1074	1082	911	743	486	318		
	CDD65	3844	38	88	190	332	490	578	609	617	461	290	109	42		
	CDH74	40853	430	874	1762	3388	5276	6852	7263	7380	4358	2203	698	369		
	CDH80	20494	128	342	766	1716	2731	3707	3985	4054	1980	781	198	106		
(14) Wind	WSAvg	7.7	7.0	7.3	8.0	8.5	8.8	9.1	9.0	8.1	6.5	6.6	6.5	6.7		
(15) Precipitation	PrecAvg	25.7	0.7	0.7	0.8	1.5	2.0	2.7	2.6	3.3	7.2	2.3	1.1	0.9		
	PrecMax	45.1	3.2	2.9	3.7	6.6	4.4	8.6	13.1	9.0	16.4	5.9	3.7	3.8		
	PrecMin	16.6	0.0	0.0	0.0	0.2	0.1	0.5	0.4	0.5	0.8	0.1	0.0	0.0		
	PrecStd	7.2	0.7	0.7	0.8	1.5	1.2	1.8	3.0	2.6	4.5	1.7	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	95.3	98.9	102.6	105.4	104.0	102.5	102.2	99.0	94.9	91.5	88.2		
		MCWB	62.0	64.4	63.8	67.9	71.6	74.0	75.1	74.4	72.9	73.0	66.2	61.9		
	2%	DB	83.8	89.2	93.3	98.4	100.2	100.4	100.2	95.3	91.0	85.7	82.4			
		MCWB	62.0	63.8	66.5	69.5	73.7	74.5	73.5	73.0	73.6	73.0	67.4	63.3		
	5%	DB	78.8	84.0	88.0	93.5	96.8	97.2	98.6	98.4	93.0	87.7	80.9	77.4		
		MCWB	60.9	63.5	66.2	69.4	73.6	73.4	73.7	72.8	73.3	72.3	66.8	62.5		
	10%	DB	73.7	79.0	83.8	89.8	93.1	95.1	96.5	95.4	89.9	84.3	77.3	73.3		
		MCWB	59.2	63.0	65.9	69.0	73.4	73.8	73.1	73.0	73.8	71.5	65.6	60.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	70.6	73.9	77.1	79.7	80.7	80.1	80.3	79.6	78.8	73.1	69.7		
		MCDB	78.9	86.0	88.3	93.0	95.9	96.2	95.5	95.9	90.1	87.8	82.7	78.8		
	2%	WB	65.7	68.1	71.0	74.6	78.0	79.0	78.9	78.8	78.2	76.6	71.0	67.3		
		MCDB	77.6	80.3	82.8	88.0	92.7	93.3	93.8	92.6	88.6	84.9	78.6	75.7		
	5%	WB	63.5	66.6	69.6	72.9	76.7	77.7	77.8	77.7	77.2	75.1	69.7	65.1		
		MCDB	73.7	78.1	80.4	85.3	89.9	91.4	91.7	90.9	87.4	83.0	76.6	73.6		
	10%	WB	61.3	64.8	68.0	71.4	75.3	76.5	76.4	76.5	76.0	73.7	68.1	63.1		
		MCDB	70.1	74.4	77.6	82.4	87.0	89.3	88.9	88.5	85.0	80.8	74.7	70.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	22.8	22.4	23.1	21.0	21.3	21.6	22.0	18.3	18.3	18.9	20.5		
		MCDBR	29.7	31.1	30.2	29.8	27.1	24.5	25.2	25.0	22.2	21.3	23.7	27.8		
		MCWBR	13.4	12.8	10.7	10.0	8.3	7.1	7.4	6.9	6.9	7.8	10.7	13.2		
		MCDBR	24.1	24.5	22.3	23.9	21.7	21.6	21.2	20.8	18.4	17.1	18.1	21.6		
	5% WB	MCWBR	12.4	11.5	9.5	9.4	8.4	7.7	7.7	7.6	7.2	7.8	9.7	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.349	0.386	0.410	0.443	0.409	0.403	0.395	0.409	0.376	0.365	0.338		
	taud		2.514	2.442	2.348	2.304	2.251	2.404	2.444	2.467	2.399	2.476	2.439	2.490		
	Ebn at Noon		295	295	289	283	272	279	280	283	278	283	279	286		
	Edh at Noon		30	34	39	42	44	38	36	36	37	33	32	29		
(44) All-Sky Solar Radiation	RadAvg		1166	1435	1719	1938	1999	2151	2098	2058	1681	1511	1248	1094		
	RadStd		122	187	178	155	113	121	159	106	154	153	145	121		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-1.78	-1.01	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page